Wet antenna attenuation in commercial microwave links

Where the research stands, explained from the beginning

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This note presents Part 1 of our project to a reader who knows the topic only by name. It describes the problem, the data, the dry reference level (the baseline), the wet antenna model and the checks that support it. Methods that were tried and not adopted are left out. The complete model, with every parameter and its uncertainty, is given in Section 8 (Figure 8).

1 The problem

1.1 Microwave links as rain sensors

Mobile phone networks connect their towers with point-to-point radio links called commercial microwave links (CMLs). Each link is a beam between two dish antennas on rooftops, typically a few kilometres apart, at frequencies of tens of GHz. The radios record the transmitted signal level (TSL) and the received signal level (RSL), both in dBm. Their difference is the attenuation along the path,

$$A(t) = \mathrm{TSL}(t) - \mathrm{RSL}(t), \tag{1}$$

in dB. Raindrops absorb and scatter microwaves, so A rises when it rains on the path. Because the networks are dense and already exist, they can serve as rain sensors, which is the long-term aim of the lab. The standard model for the rain part is a power law integrated along the path,

$$A_{\mathrm{rain}}(t) = \int_0^L a\, R(l,t)^b \, dl \approx a\, \bar{R}(t)^b \, L, \tag{2}$$

where R is the rain rate in mm/h, $\bar{R}$ its mean along the path, L the link length in km, and a and b coefficients that depend on frequency and polarisation.

1.2 The wet antenna

Rain also wets the covers of the two antennas (the radomes). A water film on a radome attenuates the signal. This is wet antenna attenuation (WAA), written here as W, in dB. WAA is local to the antennas, so it does not grow with the link length. It also has memory: the film builds up within minutes after rain starts and dries within minutes after rain stops.

Figure 1 shows the consequence. On a 0.36 km link the model attributes most of the measured attenuation to WAA. On a 4.13 km link WAA is a smaller share during heavy rain, and after the rain stops it is the only part left. A rain estimate that ignores WAA therefore overestimates rain, and the error is largest on short links.

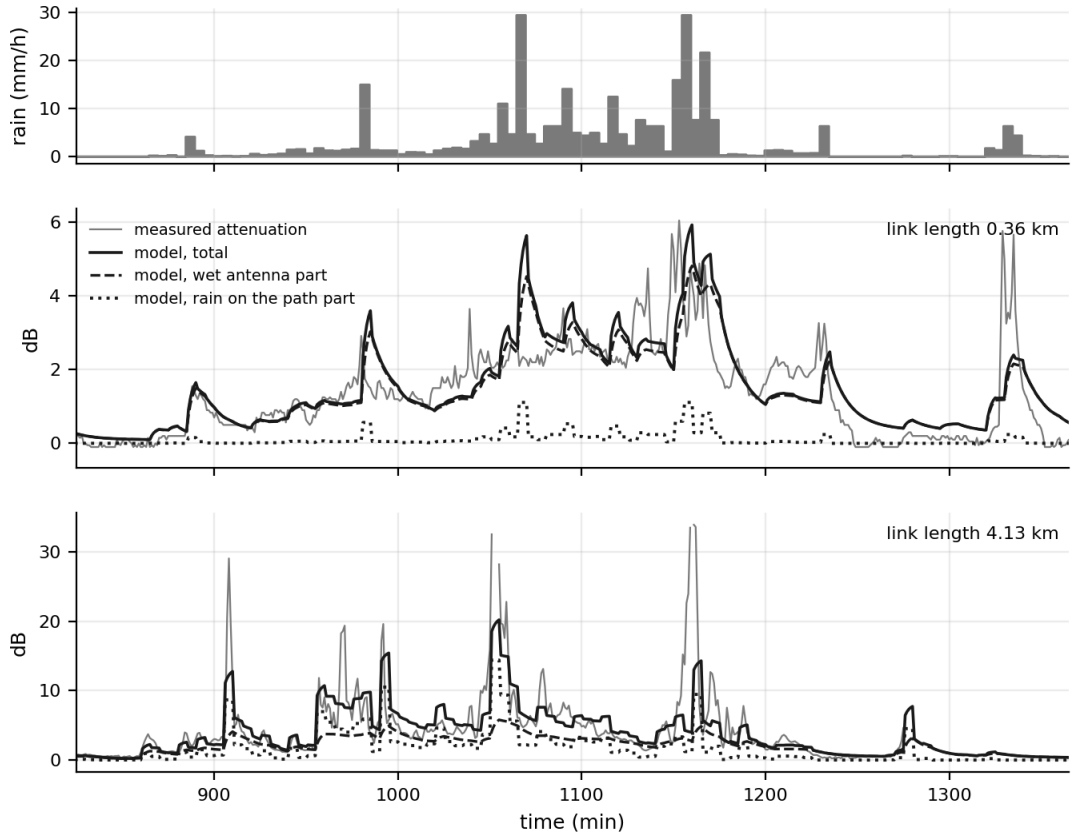


Figure 1: Real data. One storm of the training set seen by a short (0.36 km) and a long (4.13 km) link at 28 GHz, one value per minute. Top: the rain over the short link. Thin grey line: the measured attenuation $A - c$. Solid: the model. Dashed: the WAA part of the model, W. Dotted: the path part of the model, $L\,P$.

1.3 Our goal

Part 1 of the project, the subject of this note, derives an equation for $W(t)$ from data: its size, its dependence on the rain and its growth and decay in time. Part 2 will use this equation to study how the rain part depends on the link length. Part 1 follows one rule. WAA is never computed by subtracting a power-law rain term from the measurement, because every error of that formula would then be counted as WAA. The data themselves must separate the two parts.

2 The data

We use OpenMRG, an open dataset from Gothenburg, Sweden, that covers 1 June to 31 August 2015 [1]. It contains three kinds of measurement.

- Links. 728 sub-links (364 links, each measured in both directions), sampled every 10 s. RSL is quantised to 0.3 dB and TSL to 1 dB. The frequencies range from 7.5 to 38.7 GHz.
- Weather radar. One rain map every 5 min on a grid of about 2 km by 2 km. It covers the whole path of every link, but its scale error differs from storm to storm.
- Rain gauges. Ten city gauges with 1 min resolution. They measure rain on the ground accurately, but only at their own location.

A separate data pipeline cleaned and synchronised the link data, averaged them to 1 min and 5 min values, and marked rain events and quality problems. We kept the 28, 32 and 38 GHz bands, which hold 660 of the 728 sub-links (91 %). The other 68 sub-links belong to links of 4 km or more, so their antenna part cannot be separated from the path part by comparing lengths.

The data were split before any modelling (Table 1). Whole rain events were assigned to one of three sets: the training set (TRAIN) for fitting, the validation set (VAL) for choosing between model versions, and the test set (TEST) for one final check. In addition, 71 of the 364 links (20 %) were held out as unseen links. TEST was read once, after the model had been frozen (Section 6).

Table 1: The split of the data. A link-event is one sub-link during one rain event. Two of the 53 TRAIN events contain no usable link-event.

Set	Events	Link-events	Use
TRAIN	53	9 780	fitting the model
VAL	18	3 250	choosing between versions
TEST	18	3 355 + 1 010	one final check (known links + unseen links)

Rain is rare in the data. Only 12 % of the minutes used have a rain rate of 0.5 mm/h or more, and only 0.4 % have 10 mm/h or more. Results for heavy rain therefore rest on few data.

3 The dry level (baseline)

Without rain a link still has an attenuation level that differs between links and drifts slowly. This dry level, the baseline, must be known for every link and every event before the rain part can be studied. From here on, A denotes the attenuation of Equation 1 minus the baseline.

3.1 Method

For each event and each sub-link the baseline is one constant value.

1. The 24 h before the event start are taken (48 h if less than 1 h of dry data remains).
2. Only 5 min slots that radar and gauges mark as dry, and that start at least 1 h after the last rain, are kept. This excludes the drying after earlier rain.
3. The baseline is the median attenuation over these slots.

Links with a large daily cycle of the dry level, or with erratic behaviour, are flagged and excluded from the WAA study.

3.2 Validation

In real rain the true baseline is unknown, so the method was tested where the truth is known. Fake events were placed inside dry periods, where the correct WAA is zero, and the baseline error was measured. For 6 h events the median error was 0.00 dB, and the 10th and 90th percentiles were -0.21 and $+0.22$ dB. The error has no bias and therefore averages out over many events. Each sub-link keeps its own measured error, and sub-links with an error above 0.3

dB are excluded. A second test checked that the baseline does not absorb the drying after rain. One hour after real rain the level was 0.04 dB above that of a dry control, and the difference fell to zero within 12 h, as expected for a drying antenna.

3.3 A side finding: dew

The same test showed that on humid nights the attenuation can rise without rain. This is consistent with dew on the radomes, that is, WAA without rain. Such nights are flagged. Dew was not used to build the model, but it served later as an independent check (Section 7).

4 The wet antenna model

4.1 What a link measures

During a rain event, the measured attenuation of a link is written as

$$A(t) - c = W(t) + L\,P(\bar{R}(t)). \tag{3}$$

Here c is a small residual offset, defined as the median of A over the dry minutes from 120 to 30 min before the first rain on the link. W is the WAA, and $P(R)$ is the specific path attenuation, the loss per km caused by rain on the path, in dB/km. P is not fixed in advance but learned from the data. Table 2 lists the assumptions behind Equation 3, the physical reason for each, and the test that checked it.

Table 2: The assumptions of the model.

Assumption	Physical reason	Checked by
The two parts add up	They are separate losses in series	the length test (Section 4.2)
W does not depend on L	Every link has two antennas, whatever its length	three tests (Section 5.3)
The path part is L times a loss per km	Each km of path in the same rain causes the same loss	a quadratic term in L differs from zero in 4 of 15 rain groups only
The radar mean along the path, scaled by the gauges, is the right rain input	The radar covers the path, the gauges set its scale	the scaling closes the gap between TRAIN and VAL (Section 4.4)
Rain drives the wetting	Water on the radome comes from the rain	dew is excluded from the model on purpose

The third assumption is an approximation. Rain is not uniform along a path, so $L\,P(\bar{R})$ differs from the true integral in general. It is exact when P is linear in R, because then $\int c\,R(l)\,dl = c\,L\,\bar{R}$ for any rain pattern. The data showed that P is close to linear (Section 4.5), so the error caused by uneven rain is expected to be small. The general dependence of the rain loss on length is the subject of Part 2.

4.2 Separating antenna and path by link length

Aim. This step estimates, from the data alone, how much of the measured attenuation comes from the antennas and how much from the rain on the path. It gives the size of WAA and its

growth with rain without any rain formula. The later model is built on this result and checked against it.

The problem. For one link at one time, Equation 3 reads $A - c = W + L\,P$ with two unknowns. Many pairs fit the same measurement. A 1 km link that measures 3.2 dB could have 2.6 dB from the antennas and 0.6 dB from the path, or 1.0 dB and 2.2 dB. One link alone cannot separate them.

The idea. The two parts depend differently on length. The antenna part is the same for every link, because every link has two antennas. The path part grows with length. Between links of different length in the same rain, any difference therefore comes from the path. Table 3 gives an example with three links.

Table 3: An illustrative example of three links in the same rain.

L (km)	1	3	5
$A - c$ (dB)	3.2	4.4	5.6

Each additional km adds 0.6 dB, so $P = 0.6$ dB/km. Removing the path part from the 1 km link leaves $3.2 - 0.6 = 2.6$ dB, which is the antenna part. The same 2.6 dB remains on the other two links.

Method. In real data many links rarely see the same rain at the same time. Instead, all moments of steady rain whose path rain falls in the same range, for example 2 to 5 mm/h, are collected from all links and events. This set is a rain group. Each link contributes one point per event, the median of its $A - c$ in that group. A straight line is fitted through the points with a robust (Huber) fit, which limits the influence of outliers,

$$A - c = I + s\,L. \tag{4}$$

The intercept at $L = 0$, a link without a path, is the antenna part, $I = W$. The slope is the loss per km, $s = P$. This procedure is called the length test.

Output. Each rain group gives one value of W and one value of P. Several rain groups give several points of the curves $W(R)$ and $P(R)$ (Figure 2). These measured curves are the reference that the full model must reproduce.

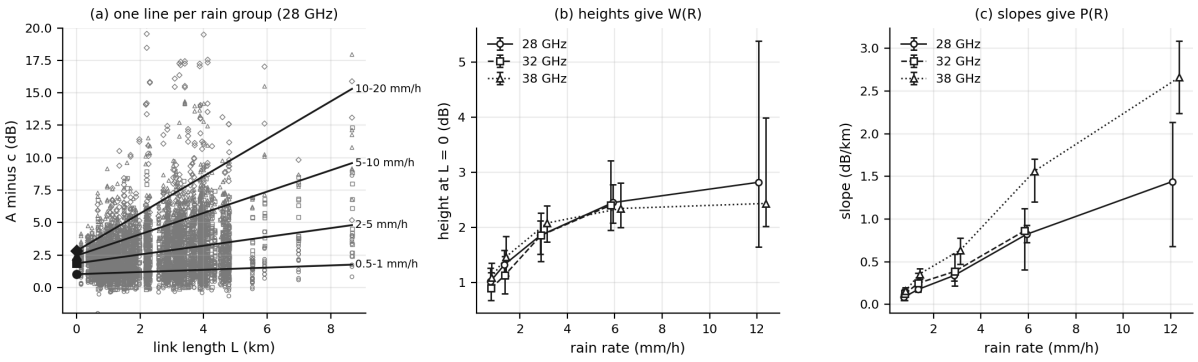


Figure 2: Real data (TRAIN). The length test. (a) Each point is one link in one event at 28 GHz, grouped by rain rate, with a straight line per group (marker shapes show the group). (b) The intercepts at $L = 0$ form the WAA curve $W(R)$. (c) The slopes form the path attenuation curve $P(R)$. Bars are 90 % intervals, line styles and markers show the band.

On TRAIN the intercept rose from about 1.0 dB at 0.5 to 1 mm/h to 2.4 to 2.8 dB at 5 to 20 mm/h and then grew only slowly, while the slope grew roughly in proportion to R. The test also showed why short links alone are not sufficient. When all links are short, a small path loss and a small additional antenna loss have the same effect. On links below 1 km the fitted path loss halved when links up to 3 km were added. The final model is therefore fitted on links of all lengths together.

4.3 Memory: wetting and drying

The length test gives a mean level, but the water on the radome does not follow the rain instantly. We model it as a store that relaxes toward a target set by the current rain,

$$\frac{dW}{dt} = \frac{u(t) - W}{\tau}, \qquad u(t) = a_{\text{link}}\, F\big(R(t)\big). \tag{5}$$

$F(R)$ is the steady-state WAA in constant rain R, and a_{link} is a dimensionless factor per link, the link factor, because some antennas collect more water than others. When W is below its target the store fills (wetting), and when it is above, it empties (drying). Wetting and drying have different time constants. In addition, a small part of the water remains for hours, so the store is split into a fast and a slow part,

$$\frac{dW_f}{dt} = \frac{0.72\,u - W_f}{\tau_f}, \qquad \tau_f = \begin{cases} \tau_{\text{wet}} = 2.6 \text{ min} & \text{if } 0.72\,u > W_f \text{ (wetting)} \\ \tau_{\text{dry}} = 10.3 \text{ min} & \text{if } 0.72\,u \le W_f \text{ (drying)} \end{cases} \tag{6}$$

$$\frac{dW_s}{dt} = \frac{0.28\,u - W_s}{240 \text{ min}}, \qquad W = W_f + W_s. \tag{7}$$

The condition on τ_f separates wetting from drying. When the rain sets a target above the water on the radome, the fast store fills with the short time constant. When the rain weakens or stops, the target falls below the stored water and the store empties with the long time constant. For a step in the rain the two cases give

$$\text{wetting: } W_f(t) = 0.72\,u\,(1 - e^{-t/2.6}), \qquad \text{drying: } W_f(t) = W_0\, e^{-t/10.3}, \tag{8}$$

with t in minutes and W_0 the stored water when the rain stops. After 2.6 min of wetting the fast store has covered 63 % of the way to its target, and after 10.3 min of drying 37 % of its water remains.

The two time constants were measured with the rain gauges, which record the start and end of rain to the minute, against the rise and fall of the links. On TRAIN they are 2.6 [2.4, 3.1] min for wetting and 10.3 [9.2, 11.8] min for drying (90 % intervals), and on VAL 2.0 [1.5, 3.2] and 8.8 [5.8, 13.1] min. The radar, with its 5 min maps, cannot resolve them. On fake data, a 5 min noisy input alone made the wetting 2.1 times slower. These values, the fast share of 0.72 and the slow time constant of 240 min, were therefore held fixed in the final fit. Figure 3 shows the response to 1 h of steady rain.

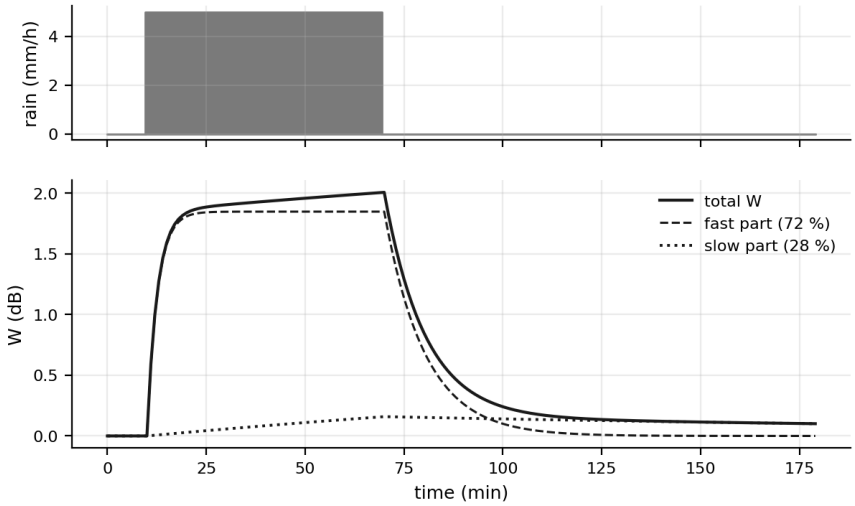


Figure 3: Illustration computed from the equation (not measured data). 60 min of 5 mm/h rain on a 28 GHz link with $a_{\mathrm{link}} = 1$. The fast store fills within minutes and empties within about 30 min, and the slow store remains for hours.

4.4 Correcting the rain input with the gauges

The model needs the rain along the path of every link, every minute. The ten gauges are accurate but measure at ten points, while the links form hundreds of paths. Only the radar covers every path. The two are therefore combined: the radar gives where and when it rains on each path, and the gauges give how much. For each event with at least 1 mm of gauge rain per gauge link, the radar is divided by one factor per event,

$$R(t) = \frac{R_{\mathrm{radar}}(t)}{g_e}, \qquad g_e = \frac{\text{radar rain in event } e}{\text{gauge rain in event } e}, \tag{9}$$

and $g_e = 1$ otherwise. In the large VAL storms the radar measured about 1.7 times the gauge rain, against about 0.9 times on TRAIN. Before the correction the model fitted VAL worse than TRAIN, and the VAL length test passed in at most 1 of 8 rain groups. After the correction it passed in 5 of 6. The difference between the sets was therefore in the rain input, not in the antennas. A test of whether the radar error depends on the rain rate, which would require a curve instead of one factor, found no such dependence. The apparent dependence reversed direction when the data were binned by the radar instead of the gauge, which is the signature of regression to the mean between two noisy measurements.

4.5 The equation and its parameters

All parts were first fitted together on TRAIN with F and P as free curves, that is, as values at fixed rain rates between 0.05 and 40 mm/h joined by straight lines, with no formula assumed. Three results followed. First, one shape of F for the three bands, scaled by one level per band, predicted VAL as well as separate shapes (root mean square error, RMSE, 0.3875 against 0.3878 dB), although TRAIN alone favoured separate shapes. Second, F rises steeply in light rain and more slowly at higher rates. Third, P is close to linear in R.

Closed formulas were then fitted to these free points and compared by the corrected Akaike information criterion (AICc), which weighs the fit against the number of parameters. The candidates were physical forms (curves that saturate at a maximum, and capped power laws) and

forms proposed by symbolic regression with the PySR library. The simplest form that describes F as well as the free points is a power of R close to a square root, proposed by PySR. It was inserted into the full model, which was then refitted. The result is

$$u = a_{\text{link}}\, F_{\text{band}}(R), \qquad F_{\text{band}}(R) = A_{\text{band}} \left(\frac{R}{5\ \text{mm/h}} \right)^{b}, \qquad P(R) = c_{\text{band}}\, R, \tag{10}$$

with $b = 0.54\,[0.47,\, 0.63]$ (90 % interval from refitting on 40 resampled sets of whole TRAIN events). Values within this interval cannot be distinguished. The interval contains 0.5, so the steady-state WAA grows approximately as the square root of the rain rate, and it excludes a linear dependence ($b = 1$). A_{band} is the steady-state WAA at 5 mm/h. Table 4 gives all values and Figure 4 the curves. The formula has no maximum, so the data do not show whether WAA saturates. Above 10 mm/h it rests on 0.4 % of the minutes, and it is not valid above 40 mm/h. Section 6 describes its test on TEST.

The linear form of P is consistent with the power law $k = \alpha R^{\beta}$ of the ITU-R P.838 recommendation. At 28 to 38 GHz its exponent is $\beta = 0.86$ to 0.92, so the power law is itself nearly linear in this range. No shape was imposed. P was learned as free points, and a power law fitted to these points gave an exponent of 0.97, with a quadratic term of zero. The linear form was selected because it has fewer parameters and fits equally well. Its size is 0.52 to 0.74 times the ITU-R value. Whether this reflects a remaining error in the radar rain or an effect of uneven rain along the path is a question for Part 2.

Table 4: The fitted parameters of the final equation, with 90 % intervals from 40 event-bootstrap refits on TRAIN. A_{band} is the steady-state WAA at 5 mm/h, b the exponent of F and c_{band} the path attenuation per km per mm/h. All were recovered on fake data (Section 5.1).

	28 GHz	32 GHz	38 GHz
A_{band} (dB)	$2.57\,[2.00, 3.21]$	$2.36\,[1.70, 3.06]$	$2.77\,[2.25, 3.40]$
c_{band} (dB km^{-1} per mm/h)	$0.105\,[0.058, 0.138]$	$0.141\,[0.047, 0.190]$	$0.194\,[0.111, 0.245]$
b (all bands): $0.54\,[0.47, 0.63]$	a_{link}: 10th to 90th percentile of 206 links, 0.77 to 1.24		

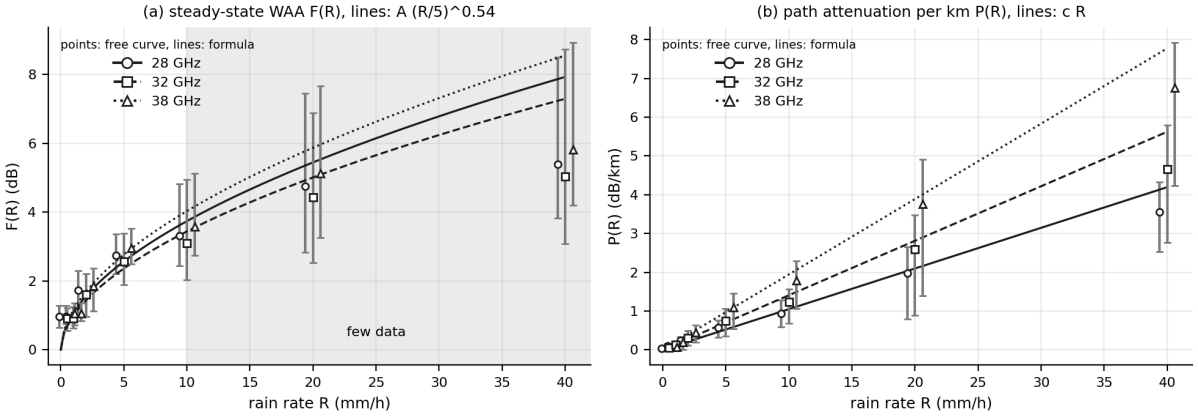


Figure 4: Real data (TRAIN). The learned curves with 90 % intervals, on linear axes. (a) Points: the free curve of F. Lines: the final formula $A_{\text{band}}(R/5)^{0.54}$, one line style per band. F is about 1 dB at 0.5 to 1 mm/h and 2.4 to 2.8 dB at 5 mm/h, and is poorly determined above 10 mm/h (shaded). (b) The path attenuation per km is close to linear in each band.

5 Evidence for the results

A fitted number is reported as a result only if it passed checks whose pass criteria were written down before the check was run. Three kinds of check were used.

5.1 Fake data

Artificial data were generated from the fitted model, with realistic noise, on the real links and the real rain, and the model was fitted again. All 62 parameters of the free curves were recovered within 20 % of their true values in each of three fake data sets, 61 of them within 10 %. The 9 parameters of the final equation were also recovered within 20 %. The method can therefore measure them. Eight parameters, the path attenuation below 1 mm/h, are too close to zero to be distinguished from it and are not reported as results.

5.2 Unseen events

Every change to the structure of the model was judged by its prediction error on events left out of the fit. A more complex version was kept only if it predicted unseen events better and its additional parameters were recovered on fake data. The choice among closed formulas used AICc on the learned points, and the chosen formula was then checked on unseen events.

5.3 Independence of the WAA from the link length

The central assumption of the model is that W does not depend on L. It was tested in three independent ways.

- The link factor a_{link} does not depend on the length (Figure 5). The Spearman rank correlation is $+0.05$, with a 90 % interval of -0.07 to $+0.17$.
- The length test of Section 4.2 was repeated with links shorter than 3 km only, and with links longer than 1.5 km only. If WAA does not depend on length, both versions give the same intercept at $L = 0$. They agreed in 13 of 15 rain groups.
- In 9 081 periods after rain the radar showed no rain on any pixel along the path. There the path part is zero, so the measurement is WAA alone, obtained without any rain model. The measured WAA and the model error were the same for short, medium and long links, with a median model error of -0.18 dB at 10 min in every length class (Figure 6).

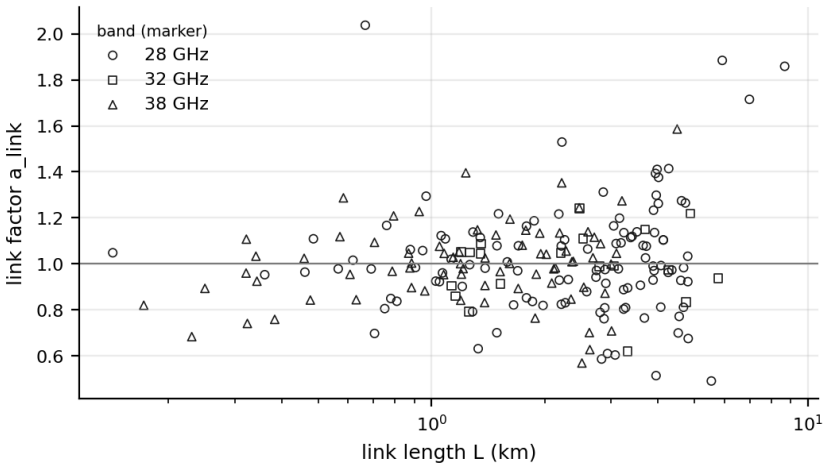


Figure 5: Real data. The link factor a_{link} of each of the 206 TRAIN links against its length. Markers show the band. There is no trend.

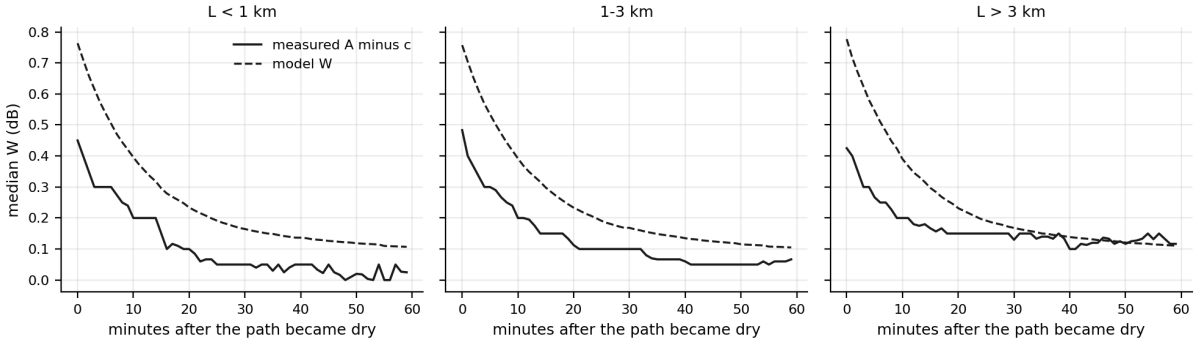


Figure 6: Real data. WAA measured after rain, when no rain remains on the path (solid), against the model (dashed), for three length classes (medians over the periods). The three classes agree. The model starts too high, which is discussed in Section 7.

5.4 Summary of the results

Table 5: The results and the evidence for each.

Result	Evidence
WAA does not depend on the link length	three independent tests (Section 5.3)
WAA has memory: wetting time constant 2.6 min, drying time constant 10.3 min, and 28 % of the water in a slow store	gauge timing, similar on TRAIN and VAL, and models with memory predict better than models without
The steady-state WAA grows approximately as the square root of the rain rate, about 1 dB at 0.5 to 1 mm/h and 2.4 to 2.8 dB at 5 mm/h	length test without a rain formula, and a formula selected among several forms, recovered on fake data, and equivalent to the free curve on TEST
The equation predicts unseen storms and unseen links better than simpler models	the final test (Section 6)
No difference between the 28, 32 and 38 GHz bands is detected	overlapping intervals of A_{band} (for example 2.00 to 3.21 against 2.25 to 3.40 dB)
One curve with a link factor describes most links	80 % of links have a_{link} between 0.77 and 1.24
The path attenuation per km is close to linear in R	free curves, the linear form selected among other forms and still passing the length test

6 The final test

All earlier checks used TRAIN and VAL. VAL had been used to choose between versions, so its errors are optimistic. TEST had not been read. Before reading it, we froze the equation, the pass rules and three simpler comparison models, and stored a cryptographic hash of each. TEST was then read once. It contains 18 storms, with 3 355 link-events on known links and 1 010 on 59 links that the model had never seen. After the read, two errors were found in the evaluation code, not in the model: storms without a usable radar-to-gauge ratio had no defined rain input. Both were fixed with the fallback $g_e = 1$ that the rule already specified, and the numbers before and after the fix are recorded.

The test does not require the true WAA, which is unknown. Every model receives only the rain (radar scaled by the gauges) and the link length, and predicts the total attenuation that the link should measure. The prediction is compared with the measurement. The comparison models were fitted on TRAIN in the same way: no WAA, a constant WAA while it rains, and a WAA without memory. The model without WAA was free to choose a larger path attenuation to explain all of A as rain on the path, and it chose 0.27 dB km^{-1} per mm/h at 28 GHz, 2.6 times the value of our equation. If a WAA part with memory predicts unseen storms better, that part describes a real effect. The split between antenna and path is checked separately, by the length test and by the dry-path periods after rain.

Table 6: The test rules, written before the test, and the results. The error is the RMSE between predicted and measured 5 min attenuation, with 90 % intervals from resampling storms.

Rule	Result	Verdict
Lower error than no WAA, a constant WAA, and a WAA without memory	0.69 dB against 0.86, 0.76 and 0.73 dB (links below 1 km: 0.56 against 0.84, 0.64 and 0.60 dB), every difference with an interval above zero	pass
Equal WAA on short and long links	dry-path error, long minus short links, −0.04 dB (−0.15 to +0.04 dB)	pass
No length class out of line	ratio of TEST to VAL error 1.12, each class within 5 % of this ratio	pass
Unseen links ($a_{\mathrm{link}} = 1$)	error 1.07 times that on known links (limit 1.20)	pass
Length test on TEST	4 of 8 rain groups agree, 6 required	fail

The failed rule locates the limit of the equation. The slopes, which measure the rain on the path, agree in 7 of 8 rain groups. The intercepts, which measure WAA, disagree in light rain (0.5 to 2 mm/h). In the test storms the intercept was 1.35 to 1.55 dB in light rain, against 0.83 to 1.11 dB in the equation. The WAA level in light rain therefore varies between storms more than the equation describes. This failure does not indicate a dependence of WAA on length, which the second rule tested and passed.

The formula for F. The equation frozen for the test still described F by free points. The formula of Section 4.5 had been fitted before the test but was not the frozen version. After the test it was evaluated on the same data, without refitting, under a rule written before the evaluation: the formula is adopted only if it gives the same five verdicts and an error within 0.01 dB of the free points. Both conditions held. The error differed by +0.0025 dB (interval −0.0065 to +0.0085 dB), and all five verdicts were the same. The formula is therefore the written form of F.

Comparison with published WAA models. After the test, and without changing our equation, we compared it with two published WAA models on the same task. This comparison was added after the freeze. The published models were fitted on TRAIN only, and no part of our equation depends on the result.

- Pastorek et al. [4] relate the WAA directly to the rain rate, $W = C\,(1 - e^{-dR^{\zeta}})$ with $d = 0.1$ and an upper limit C. The model has no memory, so the WAA disappears as soon as the rain stops. Its exponent ζ acts inside a saturating function, so it is not comparable with our b.
- Schleiss et al. [3] describe a WAA with memory. It rises toward a maximum $W_{\max}$ while it rains and decreases exponentially after the rain.

Both models were taken from the open-source package pycomlink (version 0.6.0). Its version of the Schleiss model uses the attenuation measured on the link itself, which would give the model the answer. We therefore marked a minute as wet when the radar rain on the path exceeded 0.1 mm/h, and replaced the step that uses the measurement by an exponential drying. Each model was used in two versions. In version (a) its default parameters were kept and only the path attenuation per km was fitted. For the Schleiss model this version also needs a drying time, which we could not check in the paper. We assumed it equal to the wetting time (15 min), so this version is the pycomlink defaults with an assumed drying time, not the published model. In version (b) all parameters were fitted on TRAIN, with the same link factor as our equation.

Version (b) is the fair comparison. A difference whose 90 % interval lies above zero means that our equation is better.

Table 7: TEST error of the published WAA models against our equation (RMSE of the 5 min attenuation, 90 % intervals from resampling storms). Ours: 0.695 [0.514, 0.803] dB for all links and 0.564 [0.436, 0.633] dB for links below 1 km.

Model	RMSE, all links (dB)	Minus ours (dB)	Verdict, all links and below 1 km
Schleiss (a), pycomlink defaults, assumed drying	0.859	+0.164 [+0.084, +0.286]	ours better, ours better
Schleiss (b), fitted on TRAIN	0.703	+0.009 [-0.010, +0.039]	no difference, no difference
Pastorek (a), pycomlink defaults	0.725	+0.031 [+0.017, +0.047]	ours better, ours better
Pastorek (b), fitted on TRAIN	0.720	+0.026 [+0.015, +0.038]	ours better, ours better

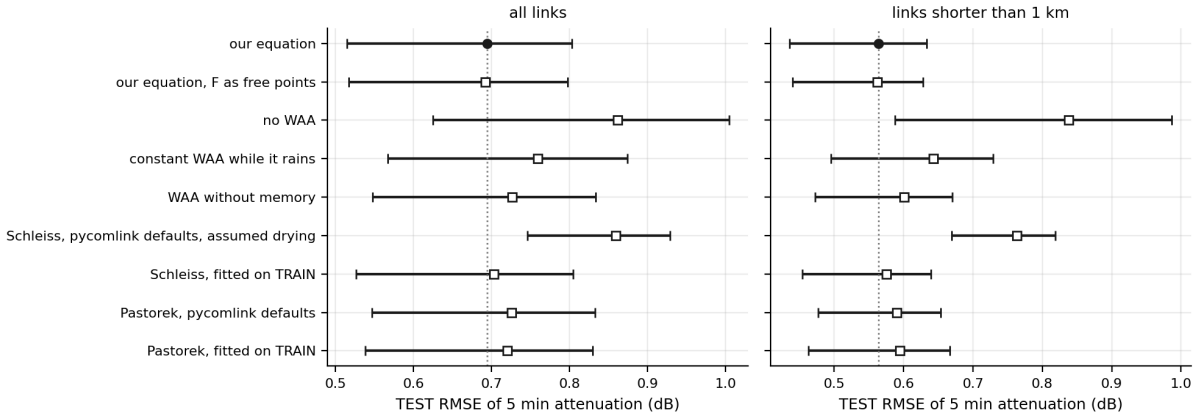


Figure 7: Real data (TEST). The error of our equation (filled circle, dotted line) against the three comparison models of the test and the two published WAA models in both versions (open squares), with 90 % intervals from resampling storms.

Our equation predicted the test storms better than both versions of the Pastorek model, by about 0.03 dB, and better than the Schleiss model with the pycomlink defaults and the assumed drying time. The Schleiss model fitted on TRAIN was as good as ours, with a difference of +0.009 dB (−0.010 to +0.039 dB), and also below 1 km. It reaches this with a different split between antenna and path. Its path attenuation per km is 1.7 to 1.8 times ours, its WAA builds up over hours (wetting time 309 min) and dries in 77 min, and after rain it holds more water than is measured (−0.25 dB at 10 min, against −0.15 dB for our equation). The Pastorek model, without memory, holds too little water after rain (+0.25 dB at 10 min).

7 Limitations

The model does not describe every effect in the data. Each effect is assigned to one of three groups.

- In the equation: the effects in Table 5.

- Observed, but outside the equation. Dew wets the same antennas: on 125 humid link-nights the median rise was 1.7 dB for about 4 h, and links with a larger a_{link} also had more dew (Spearman correlation +0.26). Directly after rain the antenna holds less water than the model (see below). These effects are described with their size and conditions but are not part of the equation.

- Not explained. The cause of the differences in a_{link} (no link property explains them). The variation of the WAA level in light rain between storms (the failed test rule). The effect of the equation on rain estimates from the links, which has not been tested. The shape of F above 10 mm/h, where data are scarce. The path attenuation below 1 mm/h.

One misfit remains. When the path becomes dry, the model holds 0.25 dB more water than is measured, and about 0.5 dB more after heavy rain (Figure 6). The possible causes were tested one by one. The measured decay rate is 0.87 times that of the model (interval 0.70 to 1.06), so the drying speed is consistent with the model, and the error lies in the amount of water left when the rain ends. The radar timing does not cause it, and the model is not too high during heavy rain. The error is larger after heavier rain and in drier air. One explanation consistent with these observations, which this work cannot confirm, is that water runs off the radome. Laboratory measurements have shown runoff directly after wetting [2]. Storms with more wind first appeared to dry faster (Spearman correlation -0.76 over 17 storms), but the effect disappeared when the preceding rain was held fixed, because the windy hours are mostly the hours after heavy showers.

A version of the model with a free drying time constant (3.7 min instead of 10.3 min) removes this misfit, but it was not adopted. With such fast drying the antenna part follows the rain almost instantly, like the path part, and the two can no longer be separated: the path attenuation halves and its interval reaches zero in every band. In addition, 3.7 min does not agree with the drying time measured with the gauges. The misfit is also a small part of the error: 2 % of the squared error on VAL and 6 % on TEST. Most of the remaining error, 61 to 64 %, occurs during rain within storms. A likely cause is that the radar resolves rain more coarsely (5 min, 2 km) than a link, but this has not been tested.

7.1 Next steps

1. Test whether removing the WAA with this equation improves the rain estimated from the links, with the gauges as reference. This step requires its own written plan and a decision about TEST, which has already been read once.

2. Part 2: the dependence of the rain loss on the link length, with the WAA removed by the frozen equation.

8 The WAA model

Figure 8 states the complete WAA model of Part 1: the inputs, the equations, the numerical scheme, every parameter with its uncertainty, and the range of validity. It is the model frozen for the test, with the formula for F adopted after the test (Section 6).

Measured quantity. $A(t) = \mathrm{TSL}(t) - \mathrm{RSL}(t) - B$ in dB, at 1 min resolution, where the baseline B is the median dry-weather attenuation over the 24 h before the event (48 h if less than 1 h of dry data remains). The offset c is the median of A over the dry minutes from 120 to 30 min before the first rain on the link (0 if fewer than 30 such minutes exist).

Inputs. Link length L (km). Band from the frequency: 28 GHz (25 to 30.5 GHz), 32 GHz (30.5 to 35 GHz), 38 GHz (35 to 45 GHz). Link factor a_{link} (dimensionless). Rain rate $R(t)$ (mm/h, one value per minute): $R = R_{\mathrm{radar}}/g_e$, with R_{radar} the radar rain averaged over the pixels along the path and held over its 5 min interval, and g_e the ratio of radar to city-gauge rain in the event over the gauges within 1 km of a path ($g_e = 1$ if the gauges caught less than 1 mm per gauge link).

Equations.

$$A(t) - c = W_f(t) + W_s(t) + L\, c_{\mathrm{band}}\, R(t-d)$$

$$\frac{dW_f}{dt} = \frac{\varphi\, u(t) - W_f}{\tau_f}, \qquad \tau_f = \begin{cases} \tau_{\mathrm{wet}} & \text{if } \varphi\, u(t) > W_f \\ \tau_{\mathrm{dry}} & \text{otherwise} \end{cases}$$

$$\frac{dW_s}{dt} = \frac{(1-\varphi)\, u(t) - W_s}{\tau_s}$$

$$u(t) = a_{\mathrm{link}}\, A_{\mathrm{band}} \left(\frac{R(t-d)}{5\ \mathrm{mm/h}} \right)^{b}$$

Initial state $W_f = W_s = 0$ in dry weather before the event. Numerical scheme: one step per minute, $W \leftarrow v + (W - v)\, e^{-\Delta t/\tau}$ with $\Delta t = 1$ min and v the target of the store (φu or $(1-\varphi)u$), output after the step.

Parameters (90 % intervals).

Parameter	Value	Source of the value and interval
A_{28}, A_{32}, A_{38} (dB)	2.57 [2.00, 3.21], 2.36 [1.70, 3.06], 2.77 [2.25, 3.40]	fitted on TRAIN, 40 event-bootstrap refits
b	0.54 [0.47, 0.63]	fitted, as above
c_{28}, c_{32}, c_{38} (dB km^{-1} per mm/h)	0.105 [0.058, 0.138], 0.141 [0.047, 0.190], 0.194 [0.111, 0.245]	fitted, as above
a_{link}	one value per TRAIN link (206 links), 10th to 90th percentile 0.77 to 1.24, 1 for a new link	fitted with the parameters above
τ_{wet}, τ_{dry} (min)	2.6 [2.4, 3.1], 10.3 [9.2, 11.8]	fixed, from the gauge-input fit on TRAIN (warm-started refits, intervals possibly too narrow)
φ	0.72 [0.60, 0.81]	fixed, as above
τ_s (min)	240	fixed at the search bound, not identified
d (min)	1	selected in the fit (integer minutes)

Validity. $0 \le R \le 40$ mm/h (above 10 mm/h based on 0.4 % of the minutes), the three bands above, link lengths 0.14 to 8.7 km (the TRAIN links), OpenMRG (Gothenburg, summer 2015). Not identified: τ_s, the path attenuation below 1 mm/h, and the level of F separately from a_{link} (the median a_{link} is close to 1 by construction).

Performance on TEST (RMSE of the 5 min attenuation, 90 % intervals). All links 0.695 [0.497, 0.795] dB, links below 1 km 0.564 [0.422, 0.630] dB. Without WAA, 0.862 [0.607, 0.998] and 0.838 [0.568, 0.977] dB.

Figure 8: The WAA model of Part 1. All values are stored in the equation file of the project, together with the per-link factors, and are computed by its reference implementation.

References

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